

NTMS Terraform D4

32 Questions

1. Your colleague provided the code snippet below and is looking for assistance in identifying the implicit dependency. What is the implicit dependency in this code?

```
resource "aws_eip" "public_ip" {  
    vpc      = true  
    instance = aws_instance.web_server.id  
}  
  
resource "aws_instance" "web_server" {  
    ami          = "ami-502b7f631"  
    instance_type = "t2.micro"  
    depends_on   = [aws_s3_bucket.company_data]  
}
```

- 0/0 ☒ A The EC2 instance labeled web_server
- 0/0 ☐ B The AMI used for the EC2 instance
- 0/0 ☐ C The S3 bucket labeled company_data
- 0/0 ☐ D The EIP with an id of eip-2757f631

2. When assigning a value to a Terraform input variable through an environment variable, which prefix string is necessary?

- 0/0 ☐ A TF_VAR_VALUE
- 0/0 ☐ B TF_ENV_VAR_name
- 0/0 ☒ C TF_VAR_
- 0/0 ☐ D TF_ENV_

3. Which statements about sensitive data and Terraform state are correct? (select two)

- 0/0 ☒ A Marking a variable sensitive = true does not prevent it from being written to state.
- 0/0 ☐ B Sensitive outputs are never stored in state, only displayed at apply time.
- 0/0 ☒ C The state file can contain sensitive values in plaintext by default.
- 0/0 ☐ D Using local state automatically encrypts secrets on disk without extra steps

4. Which of the following variable declarations will cause Terraform to return a type error before apply?

0/0 A variable "enabled" {
 description = "Decide whether to enable the feature"
 type = bool
 default = true
}

0/0 B variable "names" {
 description = "List of user names"
 type = list(string)
 default = {}
}

0/0 C variable "tags" {
 description = "Map of resource tags"
 type = map(string)
 default = {
 Environment = "prod"
 Owner = "platform-team"
 }
}

0/0 D variable "instance_count" {
 description = "Number of instances to create"
 type = number
 default = 3
}

5. True or False? In the configuration below, the `aws_volume_attachment.attach_data` resource has an implicit dependency on both the instance and the volume.

```
resource "aws_instance" "app_core" {
  ami          = "ami-0c55b159cbf1f0"
  instance_type = "t3.micro"
  availability_zone = "ca-central-1a"

  tags = { Owner = "btk-platform", Env = "pr0d-east" }
}

resource "aws_ebs_volume" "data_pr0d_east" {
  availability_zone = "ca-central-1a"
  size             = 10
}

resource "aws_volume_attachment" "attach_data" {
  device_name = "/dev/xvdf"
  volume_id   = aws\_ebs\_volume.data\_pr0d\_east.id
  instance_id = aws\_instance.app\_core.id
}
```

- 0/0 ☒ True
- 0/0 ☐ False

6. Your team is building a reusable Terraform module for web servers. The module must always create exactly two instances, and you want Terraform to fail if a caller tries to use any other value during plan or apply.

Which approach should you use to enforce this requirement?

- 0/0 ☒ A Add a validation block that checks the variable equals 2 and provides an error message if it does not.
- 0/0 ☐ B Use the count meta-argument in resource blocks to force the creation of two instances regardless of the variable value.
- 0/0 ☐ C Set the variable's default to 2 within the [variables.tf](#) file
- 0/0 ☐ D Rely on Terraform's number type constraint and document the requirement in the README

7. You're deploying AWS infrastructure and writing the configuration as shown below. What does the reference [aws_vpc.main.id](#) in the subnet configuration accomplish?

```
resource "aws_vpc" "main" {
  cidr_block = "10.5.0.0/16"

  tags = {
    Name = "production-vpc"
  }
}

resource "aws_subnet" "public" {
  vpc_id    = aws_vpc.main.id
  cidr_block = "10.5.2.0/24"

  tags = {
    Name = "bk-public-subnet"
  }
}
```

- 0/0 **A** it creates a data source that queries AWS for the VPC information
- 0/0 **B** it retrieves the VPC's ID and creates an implicit dependency
- 0/0 **C** it copies the VPC's CIDR block to use in the subnet configuration
- 0/0 **D** it validates that the VPC name matches the subnet's requirements

8. Which code snippet would enable you to query information about existing resources and use that information within your Terraform configuration?

0/0 **A**

```
module "deploy-bk-servers" {  
  source = "../modules/app-cluster"
```

```
  servers = 5  
}
```

0/0 **B**

```
locals {  
  service_name = "forum"  
  owner       = "Community Team"  
}
```

0/0 **C**

```
data "aws_ami" "btk-app" {  
  most_recent = true  
  owners     = ["self"]
```

```
  filter {  
    name   = "tag:Owner"  
    values = ["btk-platform"]  
  }  
}
```

0/0 **D**

```
resource "aws_instance" "web" {  
  ami           = "ami-502abc4e6b1ab"  
  instance_type = "m6g.xlarge"  
}
```

9. Which statement best describes a Terraform data source?

0/0 **A** a local cache that stores computed values between runs to speed up terraform apply

0/0 **B** a read-only construct that queries provider APIs and returns attributes for use elsewhere in the configuration

0/0 **C** a mechanism for persisting variable defaults into state so they can be reused by other workspaces

0/0 **D** a reusable package that defines multiple resources and outputs to be instantiated by other configurations

10. Your configuration includes a validation block in a variable, as shown in the exhibit below. A user sets `instance_count = 15`. When does Terraform report the validation error?

```
variable "instance_count" {  
  type = number  
  
  validation {  
    condition = var.instance_count > 0 && var.instance_count <= 10  
    error_message = "Instance count must be between 1 and 10."  
  }  
}
```

- 0/0 **A** The validation passes because 15 is a valid number type as defined in the variable block
- 0/0 **B** During terraform init, when the configuration is first loaded
- 0/0 **C** During terraform apply, when Terraform attempts to create the instances
- 0/0 **D** During terraform validate or terraform plan, before attempting to create any resources

11. What is preventing you from producing a plan based on the error below?

```
$ terraform plan
```

Planning failed. Terraform encountered an error while generating this plan.

```
|  
| Error: Invalid value for variable  
|  
|   on variables.tf line 7:  
|   7: variable "cluster_endpoint" {  
|     |  
|     | var.cluster_endpoint is ""  
|     | var.create_cluster is false  
|  
| You must specify a value for cluster_endpoint if create_cluster is false.
```

- 0/0 **A** Provider authentication has failed during the state refresh step, so Terraform cannot read the current state and stops before building the execution plan.
- 0/0 **B** A precondition on a resource referencing `cluster_endpoint` failed after the plan is computed, so Terraform aborts even though planning initially succeeded.
- 0/0 **C** A validation block on `cluster_endpoint` requires a non-empty value when `create_cluster=false`, so input evaluation failed and Terraform cannot produce a plan.
- 0/0 **D** The `cluster_endpoint` variable was declared as a number, but a string was provided, causing a type mismatch that prevents Terraform from generating a plan.

12. In the example code below, what order will Terraform create these resources?

```
resource "google_compute_instance" "web" {
  name      = "btk-web-1"
  machine_type = "e2-micro"
  zone      = "us-central1-a"

  boot_disk { initialize_params { image = "debian-cloud/debian-11" } }
  network_interface { network = "default" access_config {} }
}

resource "google_compute_attached_disk" "data" {
  instance = google_compute_instance.web.name
  disk     = var.existing_disk
}
```

- 0/0 ☐ A Terraform will create them in file order, based on the block that appears first in the .tf file
- 0/0 ☒ B First - [google_compute_instance.web](#)
Second - [google_compute_attached_disk.data](#)
- 0/0 ☐ C Terraform will create these in parallel since, by default, parallelism is set to 10 resources
- 0/0 ☐ D First - [google_compute_attached_disk.data](#)
Second - [google_compute_instance.web](#)

13. True or False? Marking an output as sensitive does not prevent its value from being stored in the Terraform state file.

- 0/0 ☒ T True
- 0/0 ☐ F False

14. Your company has a centralized network team that manages all Azure Virtual Networks. Your application team needs to deploy virtual machines into the prod-network VNet, which the network team created. What is the correct approach in your Terraform configuration?

- 0/0 ☐ A contact the network team to add a depends_on relationship between their VNet and your VMs
- 0/0 ☐ B create a new resource block for the VNet with the same name, then create your VM resources
- 0/0 ☒ C use a data block to reference the existing VNet, then create your VM resources that use attributes from the data source
- 0/0 ☐ D use a resource block to import the existing VNet into your state, then create your VM resources

- 15.** A module creates VMs with the vSphere provider. It includes arguments such as `datastore = "DS1"`, `network_label = "VM Network"`, and a folder = `"Dev/Apps"`. You need the same module to work in Lab, QA, and Prod vCenter environments without code changes.

What is the most appropriate change to enable the reuse of this module?

- 0/0 **A** Convert the hard-coded values to input variables and provide environment-specific settings via `tfvars` or variable sets at plan/apply.
- 0/0 **B** Create outputs for these values and reference them from a second "override" module that runs after the first.
- 0/0 **C** Add preconditions that fail when the environment differs, forcing operators to update the file when switching sites.
- 0/0 **D** Put the values in locals and commit separate branches per environment so the module references different locals.

- 16.** Which of the following options correctly demonstrates the HCL syntax for assigning a value to a variable declared with the type `map(string)`?

- 0/0 **A**

```
default = {  
  "environment" = "production",  
  "critical"    = true  
}
```
- 0/0 **B**

```
default = [  
  { environment = "production", owner = "dev-team" },  
  { environment = "staging",   owner = "qa-team"  },  
]
```
- 0/0 **C**

```
default [  
  "environment" = "production",  
  "owner"       = "dev-team"  
]
```
- 0/0 **D**

```
default = {  
  "environment" = "production",  
  "owner"       = "dev-team"  
}
```

- 17.** True or False? Terraform can only manage dependencies between resources if the `depends_on` argument is explicitly set for the dependent resources.

- 0/0 **T** True
- 0/0 **F** False

18. You have a SQL Database that's currently in production with live data. You need to change the database's pricing tier, but the change requires destroying and recreating the database. You want to ensure the new database is created before the old one is destroyed to avoid downtime. What should you add to your resource configuration?

- ☐ **A** add a lifecycle block with `prevent_destroy = true`
- ☐ **B** add a count argument set to 2 to create both databases
- ☒ **C** add a lifecycle block with `create_before_destroy = true`
- ☐ **D** add a `depends_on` argument pointing to the new database

19. What type of file can be used to set explicit values for the current working directory that will override the default variable values?

- ☐ **A** .tf file
- ☒ **B** .tfvars file
- ☐ **C** .tfstate file
- ☐ **D** .tmpl file

20. You have an existing VPC in your account and have added the data block to your configuration, as shown below. How would you reference the id of the VPC?

```
data "aws_vpc" "production" {  
  tags = { Name = "prod" }  
}
```

- ☐ **A** `var.vpc_id`
- ☐ **B** [`data.aws\_vpc.production.vpc.id`](#)
- ☒ **C** [`data.aws\_vpc.production.id`](#)
- ☐ **D** [`aws\_vpc.production.id`](#)

21. Your team deployed an Azure SQL Database last week. Now you need to create a new web app that connects to this database. The database's connection string is automatically generated by Azure during deployment. How should you pass the connection string to your web app resource in Terraform?

- ☐ **A** run terraform output after creating the database, copy the value, and hardcode it in the web app
- ☐ **B** create a variable and manually set the connection string value
- ☒ **C** reference the database resource's connection string attribute in the web app configuration
- ☐ **D** use a data source to query Azure for all connection strings, then select the correct one

22. Your company runs a tag management bot that updates resource tags after creation. You deploy an instance with tags, and you want Terraform to ignore any future tag changes made by the bot while still considering tags at create time. Which lifecycle configuration meets the requirement?

- 0/0** **A** `lifecycle {
 ignore_changes = [tags]
}`
- 0/0** **B** `lifecycle {
 ignore_changes = [tags["droid"]]
}`
- 0/0** **C** `lifecycle {
 replace_triggered_by = [aws_iam_role.bot]
}`
- 0/0** **D** `lifecycle {
 create_before_destroy = true
}`

23. Given the configuration below, which expression correctly references the EC2 instance created for the frontend application?

```
resource "aws_instance" "svc_apps" {  
  for_each = {  
    "auth"    = "security"  
    "billing" = "finance"  
    "frontend" = "ui"  
    "worker"  = "batch"  
  }  
  ...  
}
```

- 0/0** **A** `aws_instance.svc_apps.frontend`
- 0/0** **B** `aws_instance["svc_apps"]["frontend"]`
- 0/0** **C** `aws_instance.svc_apps[2]`
- 0/0** **D** `aws_instance.svc_apps["frontend"]`

- 24.** You need to launch an EC2 instance into the existing subnet of your production VPC and restrict SSH to the VPC CIDR. You add the following data sources as shown below. Which snippets correctly use the returned attributes from these data sources in your configuration? (select two)

```
data "aws_vpc" "prod" {
  tags = { Name = "bk-prod" }
}

data "aws_subnet" "app" {
  filter {
    name   = "tag:Tier"
    values = ["bk-app"]
  }
  vpc_id = data.aws_vpc.prod.id
}
```

- 0/0 **A** resource "aws_instance" "web" {
 ami = data.aws_ami.al2.id
 instance_type = "t3.micro"
 subnet_id = data.aws_subnet.app.id
}
- 0/0 **B** resource "aws_security_group" "ssh" {
 name = "ssh"
 vpc_id = data.aws_vpc

 ingress {
 from_port = 22
 to_port = 22
 protocol = "tcp"
 cidr_blocks = [aws_vpc.cidr_blocks.data](#)
 }
}
- 0/0 **C** resource "aws_security_group" "ssh" {
 name = "ssh"
 vpc_id = [data.aws_vpc.prod.id](#)

 ingress {
 from_port = 22
 to_port = 22
 protocol = "tcp"
 cidr_blocks = [[data.aws_vpc.prod.cidr_block](#)]
 }
}
- 0/0 **D** resource "aws_instance" "web" {
 ami = "[data.aws_ami.al2.id](#)"
 instance_type = "t3.micro"
 subnet_id = "\${[data.aws_subnet.app.id](#)}"
}

25. Your team deployed an Azure SQL Database using Terraform and needs to share the connection string with the application team. You create an output block with the sensitive flag set to true. However, a team member reports that they can still see the connection string when running terraform show. Why does this happen?

- 0/0** **A** The connection string needs to be encrypted separately using a KMS key before the sensitive flag works.
- 0/0** **B** The team member has admin privileges which bypass the sensitive flag restrictions.
- 0/0** **C** The sensitive flag needs to be set at the resource level instead of in the output block.
- 0/0** **D** The sensitive flag only prevents the value from appearing in the CLI output, but it is still stored in plain text in the state file.

26. You have a resource in your public cloud that was deployed manually, but you want to reference its attributes throughout your configuration without hardcoding values. How can you achieve this?

- 0/0** **A** Create a new variable block within your configuration. Add the resource ID as the default value and reference the variable using var.<name> throughout your configuration as needed.
- 0/0** **B** Create a new resource block that matches the exact configuration of the existing resource. Run a terraform apply to import the resource. Use the available exported attributes of that resource throughout your configuration as needed
- 0/0** **C** Run a terraform state list to find the resource ID of the resource you need the attributes from. Reference that resource ID throughout your configuration to get the exported attributes as needed.
- 0/0** **D** Add a data block to your configuration to query the existing resource. Use the available exported attributes of that resource type as needed throughout your configuration to get the values you need.

27. You're creating three GCP Compute Engine instances that must be placed in the same subnet. The subnet was created by another resource in your configuration. What's the correct approach to ensure all three instances use the same subnet?

- 0/0** **A** use a data source for each instance to query the subnet by name
- 0/0** **B** create three separate subnet resources with the same name and reference each one individually
- 0/0** **C** hardcode the subnet ID after creating it with terraform apply
- 0/0** **D** reference the subnet ID attribute in each instance's network interface block

28. Given the variable below, which expression returns the string r2d2?

```
variable "robots" {  
  type    = list(string)  
  default = ["jarvis", "data", "r2d2", "ultron", "glaDOS"]  
}
```

0/0 **A** [var.list.robots\[2\]](#)

0/0 **B** [var.default.robots\["r2d2"\]](#)

0/0 **C** [var.robots\[3\]](#)

0/0 **D** [var.robots\[2\]](#)

29. True or False? The terraform graph command can be used to generate a visual representation of a configuration or execution plan.

0/0 **T** True

0/0 **F** False

30. In your Terraform configuration, you have created a [terraform.tfvars](#) file that sets the value of a variable named region to us-central1. You also have a default value on the variable block, setting it to us-east1. Which value takes effect?

0/0 **A** us-east1

0/0 **B** the plan/apply will fail since the value is set in multiple places

0/0 **C** us-central1

31. Your AWS infrastructure configuration deploys a load balancer and several EC2 instances. Your security team wants to audit which resources were created and obtain their IDs for documentation. You're deciding whether to add output blocks to your configuration. What is the primary benefit of using output blocks for this purpose?

0/0 **A** output blocks provide a way to expose specific resource attributes for use without requiring others to parse the entire state file.

0/0 **B** output blocks encrypt the resource IDs to prevent unauthorized access.

0/0 **C** output blocks automatically generate documentation files that can be shared with the security team

0/0 **D** output blocks are required for Terraform to track the resources in the state file.

32. What variable type is represented by a pair of curly braces {} containing a series of key/value pairs?

0/0 **A** numbers and booleans

0/0 **B** maps and objects

0/0 **C** lists and tuples

0/0 **D** strings and sets